Final Project Submission

DeepCompile: AI-Driven Compiler Optimization

Roll Number: 24CSB0A06 · Date: April 19, 2026

Phase 2 Integration Report

Generated: 2026-04-19 01:17

LLM Mode: offline (rule-based)

Total runtime: 35.2 s

Executive Summary

This file is the frozen Week 8 baseline snapshot for the original 4-stage flow. Later Week 9-14 work in this repository extends this baseline with orchestration, bounded refinement, shared explainability exports, and deterministic rule-based evaluation under a revised source-to-source scope.

The 4-stage pipeline (Analysis -> Optimization -> Verification -> Security) was run on **8** benchmark test cases (2 skipped because the files were not present at report-generation time).

| Metric | Value |

|--------|-------|

| Files run | 6 |

| Pass rate (success) | 83.3% (5/6) |

| Rollbacks | 1 |

| Partial completions | 0 |

| Failed | 0 |

| Avg analysis findings per file | 0.7 |

| New security vulns introduced | 0 |

| Security rollbacks triggered | 0 |

Per-Agent Performance

Analysis Agent

- Total findings across all files: **4**
- High-severity findings: **4**
- Average findings per file: **0.7**
- Top bug categories detected (by filename):

- `uninitialized_variable`: 1 file(s)
- `buffer_overflow`: 1 file(s)
- `unsafe_strcpy`: 1 file(s)
- `memory_leak`: 1 file(s)
- `use_after_free`: 1 file(s)
- `heap_overflow`: 1 file(s)

Optimization Agent

- Files with transformations applied: **6** / 6
- Total transformations: **5**

Verification Agent

- Verification `PASS`: 5
- Verification `FAIL`: 0
- Verification `ROLLBACK`: 1
- Verification `skipped`: 0
- Differential tests passed: **4** / 6

Security Agent

- Total vulnerabilities found (optimized code): **3**
- New vulnerabilities introduced by optimization: **0**
- Security rollbacks triggered: **0**
- Overall risk `high`: 1 file(s)
- Overall risk `medium`: 2 file(s)
- Overall risk `none`: 3 file(s)
- Detection layers used:

- `heuristics`: active in 6 run(s)
- `rules`: active in 6 run(s)

Per-File Results

File	Category	Status	Findings	Opt	Ver	Sec Risk	New Vulns	Time (s)
`TC01_uninit_arithmetic.cpp`	Uninitialized variable - arithmetic	success	1 (H:1)	yes	PASS	none	0	4.37
`TC06_div_by_zero_simple.cpp`	Division by zero - simple	skipped	0 (H:0)	no	skipped	?	0	0.0

| `TC09_array_out_of_bounds.cpp` | Array out of bounds | skipped | 0 (H:0) | no | skipped | ? | 0 | 0.0 |

| `TC11_buffer_overflow_loop.cpp` | Buffer overflow - loop | success | 2 (H:2) | yes | PASS | none | 0 | 1.24 |

| `TC12_strcpy_overflow.cpp` | strcpy buffer overflow | success | 0 (H:0) | yes | PASS | medium | 0 | 14.21 |

| `TC14_memory_leak.cpp` | Memory leak | success | 1 (H:1) | yes | PASS | none | 0 | 1.23 |

| `TC16_use_after_free.cpp` | Use-after-free | rollback | 0 (H:0) | yes | ROLLBACK | high | 0 | 1.25 |

| `TC19_heap_overflow.cpp` | Heap overflow | success | 0 (H:0) | yes | PASS | medium | 0 | 12.91 |

Failure Analysis

`TC16_use_after_free.cpp`

- Status: `rollback`
- Verification result: `ROLLBACK`

Conclusion and Week 9 Readiness

Phase 2 Status: READY

The core 4-stage pipeline is functional. Pass rate is **83.3%** with **1** rollback. The Security Agent detected **0** newly introduced vulnerabilities across the benchmark cases captured in this baseline snapshot.

Ready for Week 9: Self-refinement mechanism and Orchestrator Agent.

Scope Note for Later Weeks

The revised implementation path now present in the repository uses:

- source-to-source optimization as the formal delivery scope
- deterministic `--no-llm` execution as the preferred evidence path
- no required LLVM/Clang or `g++` integration for the revised submission target

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Generated by `evaluation/generate_report.py` and later annotated for the revised Week 9-14 scope.